Network Manager Release Notes

Version 7.0.3.0

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1: Introduction

This release of Network Manager Telecom is compatible with IQGeo Platform 7.0.

This document provides a summary of the major changes in this release of Network Manager Telecom.

1.1 Overview

This release of Network Manager introduces the following new functionality.

Support for copper and coaxial networks

Network Manager Telecom 3.0 features support for copper and coaxial network technology in addition to fiber optics. The concept of *Technology* is introduced, which allows other network technologies to be modeled and managed. To that end, new data model templates are available for installation providing cable types and the typical equipment for these technologies.

The majority of the operations involved in modeling and managing networks can be performed using the same functions independent of the underlying technology.

For both technologies, though, specific functions have been introduced.

For copper, these are:

- A new trace type of Effective Working Length (EWL), which will trace a copper cable to a specified distance, upstream or downstream.
- A new type of report Loop Makeup Report is available from the context menu in the Equipment tree.

For coax, these are:

- Possibility to add and display Data Blocks for specific coaxial equipment, presenting attribute values of the equipment.
- The presence of a secondary (offset) geometry for coaxial cables and equipment directly displayed on the map.

Support for Line of Count (LOC)

Network Manager Telecom now offers the possibility to assign Line of Count information (name-status) to a range of strands in a cable, i.e. copper pairs or fibers).

Line of count (also known as *cable count* or *pair ranges*) is a common methodology for pair assignments in copper networks (particularly in North America). The approach is also applied to fiber distribution networks as well as it is well suited to large hierarchical networks.

Support for cable to enclosure relationships

It is now possible to create a relationship between a specific cable and an enclosure, for example splice closure, rack, shelf, etc. within a structure.

Textual filtering

On the Equipment and Cables tree view as well as in dialogs where equipment or cables need to be selected from a list, a textual filtering feature is now available.

Configuration pages updates

Various changes to the Configuration pages to support the new features listed above.

1.2 Intended audience

This document is intended for Administrators and Developers with a detailed technical knowledge of IQGeo Platform and Network ManagerTelecom.

2: Documentation

The documentation set has been updated to reflect the new features described in the sections below.

The documentation also uses a responsive layout that adapts to the layout of the users' device and allows users to search across guides.

All documentation can be found in the *public/doc* directory.

2.1 Comms Data Interchange Format (CDIF) Documentation

Detailed information about the CDIF format can be found in the document *CDIF Format.PDF* (v. **1.1**) in the *public/doc* directory.

The document contains new information on the following topics:

- Optional equipment fields on segment objects
- Technical notes for equipment and cables
- Optional **offset_geom** field on cables
- Specifications for the **Line of Count** functionality

3: Installation

The following changes relate to installation and upgrade.

3.1 Prerequisites

Component	Version	Comments
IQGeo Platform	7.0	Mandatory
IQGeo Anywhere	7.0 (or higher)	Optional

All components must include any released patches.

3.2 Installing the software

1. Uncompress the software download and copy the **comms** folder into:

   ```
   <myWorld_install_dir>/WebApps/myworldapp/modules
   ```

2. Download the module's JavaScript dependencies:

   ```
   myw_product fetch node_modules
   ```

3. Rebuild your JavaScript bundles:

   ```
   myw_product build applications
   ```

3.3 Creating a new database

If you do not already have an IQGeo database, or that database does not yet have the Network Manager tables installed, you will need to:

- Install the Network Manager tables.
- Configure the network object types for your installation.

These processes are covered in detail in the *Installation and Configuration Guide*.

3.4 Upgrading an existing database

If you already have a Network Manager database from a previous release, you will need to:

- Upgrade your database.
- Perhaps perform some post-upgrade configuration changes.

These processes are covered in detail in *Upgrading* on page 24.

4: End-user GUI changes

The following changes have been made to the end-user GUI.

4.1 Support for coax and copper network technologies in addition to fiber

> The fact that Network Manager Telecom supports the three network technologies does not imply that users are obliged to set it up in their own application. According to their needs and requirements they can decide which technology(-ies) they want to work with and install the appropriate data model templates accordingly. All explanations below assume a system with all templates installed.

4.1.1 New Layers introduced

The **Layers** that can be loaded for displaying on the map have been reorganized in function of the three network technologies that are supported. For each technology there is a separate group layer identified as ***<Technology>* Network**. Each layer, in turn, comprises two sublayers named ***<Technology>* Cables** and ***<Technology>* Equipment**. The latter lists the different pieces of equipment that have been configured for that technology and the symbols with which they are presented on the map.

You can show/hide a network layer as a whole or show/hide its sublayers individually by ticking on or off the appropriate check box.

For the coaxial technology, there is an additional layer **Coax Offset Group**, which is to render on the map sets of coaxial cables and equipment thereby observing a certain offset from the "primary" location (route). This layer also consists of a ***<...>* Cables *<...>*** and a ***<...>* Equipment *<...>*** sublayer.

Another layer that relates to the coax-based technology is the **Data Block** layer. It allows to graphically present on the map a set of relevant data concerning coaxial equipment.

Finally, there is the **Mixed Equipment** layer that presents equipment items that can be integrated in networks using two different technologies. A typical example is an optical node to which a fiber cable is connected on the *IN* side and a coaxial cable on the *OUT* side.

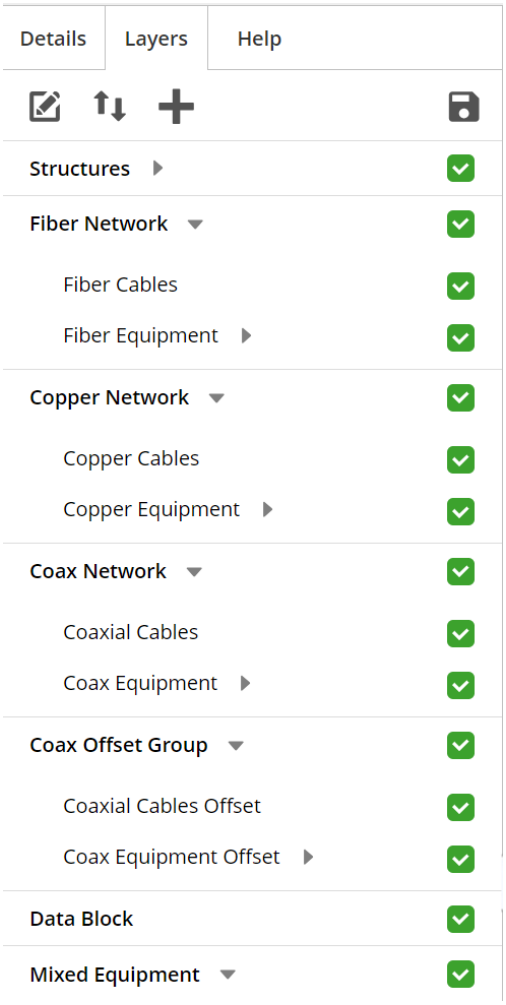


Figure 4-1: Layers technology-based structure

4.1.2 Add Object dialog updated

The **Add Object** dialog has been extended and restructured to support the different network technologies.

The **Cables** section presents the available cable objects to choose from in function of the technology: **Coaxial**, **Copper** and **Fiber**. If the user configured his own cable types, they are also listed.

The **Equipment** section comprises an extensive list of objects referring to the various equipment items relevant to the different technologies.

For copper networks typical equipment is introduced, including copper terminals, slacks, copper splices, bridge taps, build outs, repeaters, capacitors, load coils, pair gains and repeaters. All features have associated specification records.

For coaxial networks this includes all equipment typically found within an RF coaxial network such as optical nodes, amplifiers, directional couplers, taps, coaxial splitters, etc.

> It is important to know that the **Equipment** section of the list is only populated, if a compatible type of containing structure is selected.

4.1.3 Add Object - Coaxial cable

New function **Set Offset** allowing you to draw the path of the coax cable offset yourself on the map.

When you select and **Edit** a coaxial cable, the function **Update Offset** allows you to do the same for an existing cable.

4.1.4 Add Object - Data Block

With this function you can assign a Data Block to a coaxial or optical piece of equipment. The object appears on the map as a tabular overview of a set of properties of the equipment.

4.1.5 Add Object - Circuit restricted to fiber strands only

Adding a circuit through the **Add Object** dialog is only allowed for fiber cables. In the **Circuit Termination** dialog, which opens when you click the **Set Route** button from the **Circuit** dialog, you must select a fiber strand to **Set** the circuit. If you select a copper or coax strand, the **Set** button is grayed out.

> Equipment that is suited for mixed technologies, such as DLSAM or optical node, can have fiber strands on one side and copper or coax on the other.

4.1.6 Add context menu updated

The context menu that allows adding equipment to a structure from within the **Equipment** tree also allows for the different technologies when presenting the available equipment items. Depending on the type of containing structure the presented equipment items are specific to one or to several of the technologies.

From the two images shown below the first one illustrates a situation where equipment from the various technologies is accepted whereas the second one illustrates a situation where only copper-specific equipment is allowed.

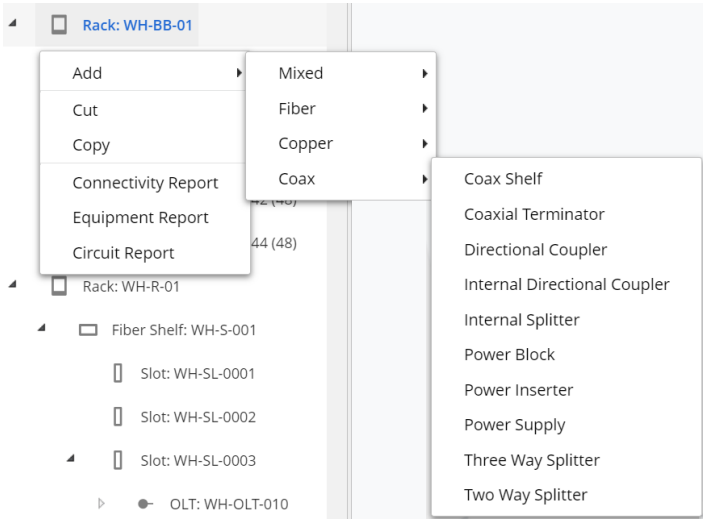


Figure 4-2: Add equipment context menu with choice between multiple technologies

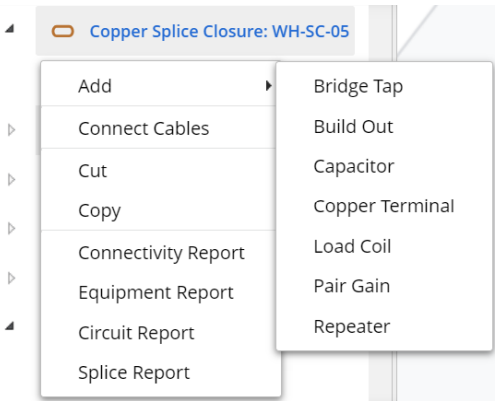


Figure 4-3: Add equipment context menu with only copper technology-related items

4.1.7 Tools palette Spec Manager updated

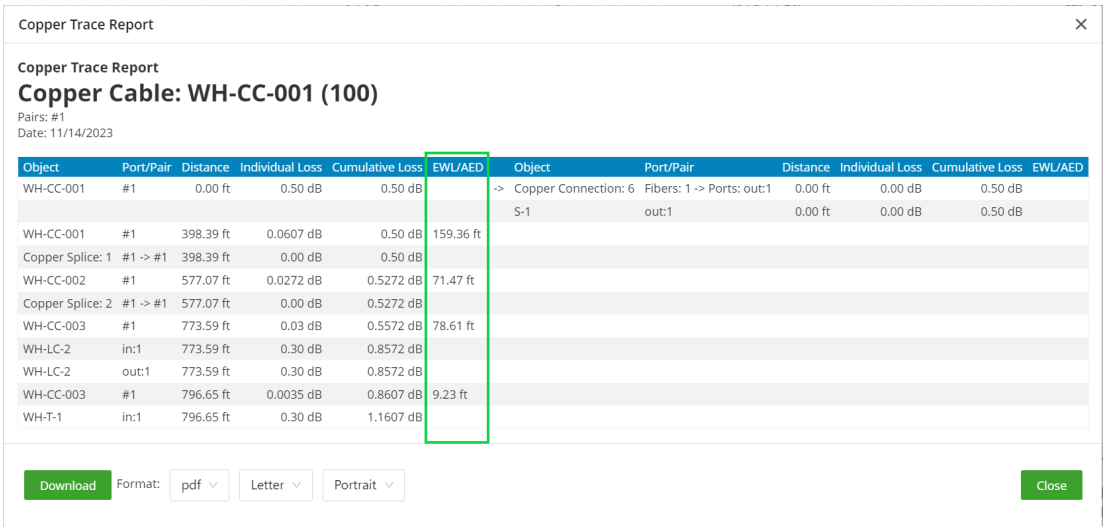
The **Equipment** section of the **Spec Manager** lists the various equipment items for the different supported technologies.

The **Cables** section contains a separate function for adding cable specifications for each of the three technologies.

4.1.8 Equivalent Working Length (EWL) data provided for copper cables

Equivalent Working Length (EWL), also referred to as Access Edge Distance (AED), can now be calculated for a copper cable for a given distance either upstream and downstream. EWL/AED information is also provided in the Trace Reports, Loop Makeup Reports and Fault Tracing Reports for copper cables.

EWL, which is based on the gauge (diameter) of the wires, is provided as a length normalized against a gauge of 26. For cables with a gauge of 26 the EWL matches the "real" length associated with the cable. Lower gauges result in EWL lengths that are gradually shorter than the associated length of the cable.



Object	Port/Pair	Distance	Individual Loss	Cumulative Loss	EWL/AED		Object	Port/Pair	Distance	Individual Loss	Cumulative Loss	EWL/AED
WH-CC-001	#1	0.00 ft	0.50 dB	0.50 dB		->	Copper Connection: 6	Fibers: 1 -> Ports: out:1	0.00 ft	0.00 dB	0.50 dB	
							S-1	out:1	0.00 ft	0.00 dB	0.50 dB	
WH-CC-001	#1	398.39 ft	0.0607 dB	0.50 dB	159.36 ft							
Copper Splice: 1	#1 -> #1	398.39 ft	0.00 dB	0.50 dB								
WH-CC-002	#1	577.07 ft	0.0272 dB	0.5272 dB	71.47 ft							
Copper Splice: 2	#1 -> #1	577.07 ft	0.00 dB	0.5272 dB								
WH-CC-003	#1	773.59 ft	0.03 dB	0.5572 dB	78.61 ft							
WH-LC-2	in:1	773.59 ft	0.30 dB	0.8572 dB								
WH-LC-2	out:1	773.59 ft	0.30 dB	0.8572 dB								
WH-CC-003	#1	796.65 ft	0.0035 dB	0.8607 dB	9.23 ft							
WH-T-1	in:1	796.65 ft	0.30 dB	1.1607 dB								

Figure 4-4: Copper Trace Report with EWL information

4.2 Support for Line of Count (LOC)

The **Line of Count** feature provides a way to model logical assignment of a *name* and *status* to connected strands (i.e. fibers or pairs) in the network.

New functionality has been introduced to support Line of Count.

The **Line of Count Section** feature relates the LOC assignment to a cable segment or to an equipment port.

4.2.1 Edit Line of Count function

To assign Line of Count to a cable segment or equipment side, use the **Edit LOC** function, which is available from the tree view context menu at the level of **Equipment** or **Cables**.

When you click it, the **Edit LOC** on the `OUT` ports of a cable, a dialog opens where you can:

- Create (a) LOC range(s) clicking the **Add** button.
- Specify the **Name** and **Status** to be assigned to the range of connected strands.
- Add multiple lines and assign them to different ranges of strands by setting the **Low** and **High** values.
- You can also re-arrange the order of assignments.
- **Riple** the assigned LOC data downstream through the different segments of the connected network. It is also possible to ripple only the status without changing the name or the range.

  When connectivity is changed the system can be configured to either automatically run a **Ripple** operation to update the pair assignments or to mark those pair assignments as invalid with an associated design check to highlight these to the end user when a design is checked.

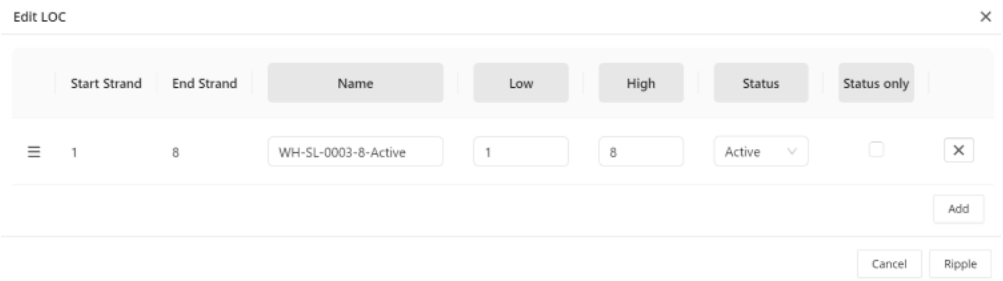


Figure 4-5: Edit LOC dialog

4.2.2 View LOC data

The **View LOC** function, also available from the tree view context menu, displays the LOC data assigned to a certain cable or cable segment.

In the tree view, for both the **Equipment** section and the **Cables** section, it is also possible to automatically show or hide the LOC data by ticking the **Show LOC** option on or off in the **Settings** menu (accessed by clicking the cog icon). The LOC information is displayed on fibers, copper pairs and ports.

4.2.3 Data Validation of LOC

The Data Validation tool includes Line of Count in its list of objects for which it can check data integrity.

4.3 New tree view functionality

4.3.1 New function Set Enclosure

To create a relationship between a specific cable (segment) and an enclosure a new function **Set Enclosure** is available in the cable tree context menu. Once established these relationships will control connectivity within the **Connect Cables** dialog by restricting connections to only those cables with the same equipment relationships, i.e. it will no longer be possible to connect fibers in cables that are not in the same enclosure. The **Connect Cables** dialog has been enhanced to include enclosure filters for the **From** and **To** panes.

Technically this is implemented with the addition of new fields **in_equipment** and **out_equipment** available for **mycom<tech>segment** objects.

This will not impact exiting data, only data where the relationship has been explicitly set.

The context menu available from within the cable tree of a structure includes a new function **Set Enclosure** for cable segments.

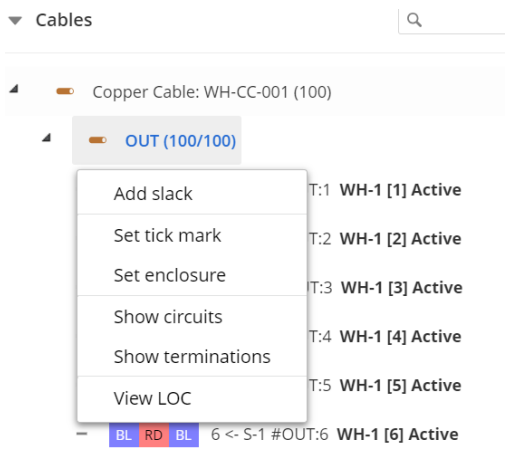


Figure 4-6: Set Enclosure for cables

It allows selecting a piece of equipment present in the structure and assign it as the enclosure of the cable segment concerned.

4.3.2 Textual filtering

The **Equipment** and **Cables** sections of the tree view in the **Details** pane provide a search field for textual filtering. You can use this feature when you want to perform a targeted search for a specific equipment item or cable. You enter the name of the object or part of it in this field so that the list only shows items that match the specified string.

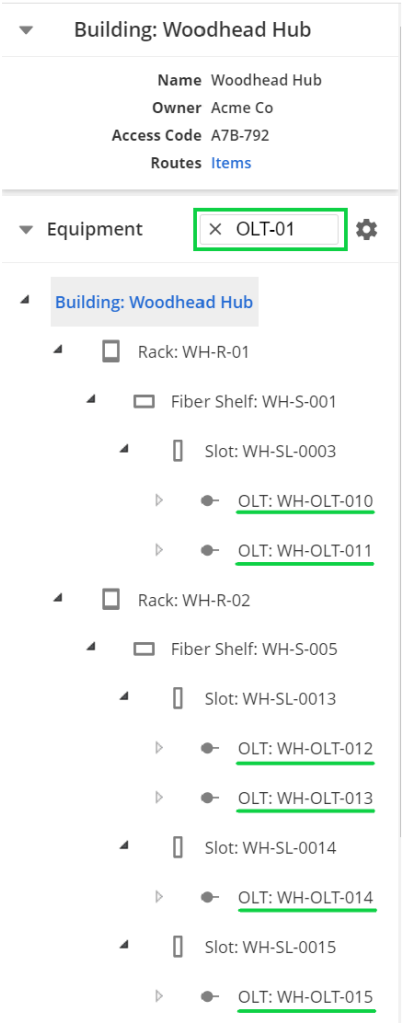


Figure 4-7: Equipment tree textual filtering

Similar search/filtering functionality is also available in various dialogs:

- the **Connect** dialog that is displayed when you use the **Connect** function to connect cables to equipment (ports).
- the **Connect Cables in...** dialog that opens when you use the **Connect Cables** function on a splicing closure to connect cables to cables (splicing) .
- the **Select Equipment** dialog that opens when you use the **Set Enclosure** function (see higher).

4.3.3 Filtering by containment on cable connection trees

The dialog that opens when you use the **Connect Cables**function from the **Equipment** tree context menu supports filtering by containing structures. Selecting a structure from the list at the bottom, automatically limits the list of cables to those that are contained by the selected structure. The filtering can be applied to the cables in the **From** pane as well as the **To** pane.

5: Administrator GUI changes

The following changes have been made to the Administrator GUI:

5.1 Networks Configuration Page

To support coax and copper network technology, two new networks have been added:

- mywcom_coax
- mywcom_copper

Both networks provide a list of technology-specific network features for which additional configuration is possible.

5.2 Layers Configuration Page

The list of available layers that can be selected for displaying in the map of Network Manager Telecom has been extended with layers grouping technology-specific equipment and cables for coax and copper networks or equipment that can be used for mixed technolgies..

The Line of Count data has also been added as a possible layer.

5.3 Features Configuration Page

The list of feature types that can be configured has been extended with network objects and their `_spec` features that are used in networks based on coax and copper technology.

5.4 Settings Configuration Page

5.4.1 Comms Settings

- **Equipment tab**

  The list of equipment feature types that can be configured has been extended with the various types of equipment needed to support the three network technologies.

- **Cables tab**

  Network cables are identified by the network technology that they are used for:

  - fiber_cable
  - copper_cable
  - coax_cable

- **Specs tab**

  The list of equipment feature types for which you can configure the **Spec Field** has been extended with the various types of equipment needed to support the three network technologies.

- **Labor Costs tab**

  The list of feature types for which you can configure the **Labor Costs Field** has been extended with the various specification feature types of equipment items needed to support the three network technologies.

5.4.2 Advanced Settings

The following setting has been added to the **Advanced** tab of the Settings configuration page:

- `mywcom.network_types`

  These settings establish the three network types based on fiber, coax and copper respectively and the characteristics of the network types.

- `mywcom.dataBlocks`

  These settings determine the coax equipment for which data blocks can be generated and the data fields they are to include.

- `mywcom.ewl`

  These settings are to establish how the gauge values of a copper cable determine its EWL.

- `mywcom.line_of_count`

  These settings are to determine whether LOC data is automatically rippled through the various segments when there is a change in the connection status of the fiber or copper pair.

6: Data Model changes

6.1 Coaxial and Copper Networks Support

As of this version Network Manager Telecom supports the creation and modeling of telecoms networks based on coax and copper technology In addition to fiber-based networks. To this end, several changes have been made to the data model.

6.1.1 New System Objects

A set of new technology-specific system objects have been added:

Model	Description
`mywcom_coax_segment`	Represents coax cable segments and their properties.
`mywcom_coax_connection`	Represents coax connections and their properties.
`mywcom_coax_slack`	Represents cable slack on coax cables.
`mywcom_data_block`	Graphical representation of a set of key features of a piece of coax equipment.
`mywcom_copper_segment`	Represents copper cable segments and their properties.
`mywcom_copper_connection`	Represents copper connections and their properties.
`mywcom_copper_slack`	Represents cable slack on copper cables.

6.1.2 New Network Definitions

Two new network definitions have been added:

Model	Description
`mywcom_coax`	Tree network for modeling coaxial wires and equipment.
`mywcom_copper`	Tree network for modeling copper pairs and equipment.

6.1.3 New Network objects

Two sets of new network objects have been added:

6.1.3.1 Coax Technology

The Coax Template Model presents the following technology-specific network object types (cable and pieces of equipment):

- `coax_cable`
- `coax_amplifier`
- `coax_shelf`
- `coax_tap`
- `coax_terminator`
- `directional_coupler`
- `inline_equalizer`
- `internal_directional_coupler`
- `internal_splitter`
- `optical_node`
- `optical_node_closure`
- `power_block`
- `power_inserter`
- `power_supply`
- `three_way_splitter`
- `two_way_splitter`
- `wifi unit`

6.1.3.2 Copper Technology

The Copper Template Model presents the following technology-specific network object types (cable and pieces of equipment):

- `copper_cable`
- `copper_bridge_tap`
- `copper_build_out`
- `copper_capacitor`
- `copper_dslam`
- `copper_load_coil`

- `copper_pair_gain`
- `copper_repeater`
- `copper_shelf`
- `copper_splice_closure`
- `copper_terminal`

6.1.4 New Specification Feature Types

New specification feature types were also associated with the new network objects mentioned above. The specification feature types have the same name as the associated network object with the addition of the suffix `_spec`.

6.1.5 Settings Changes

Several settings have been in view of the support for the three different network technologies. For details, see *Administrator GUI changes* on page 18

6.1.6 New Template Models

Two new template data models have been added that can be used as the starting point for the modeling of coax and copper networks respectively:

Model	Description
coax	Template model containing the basic objects for modeling coaxial wires and equipment.
copper	Template model containing basic objects for modeling copper pairs and equipment.

6.1.7 Existing Network Object Changes

To distinguish the network objects for the three different technologies supported by Network Manager Telecom, the prefix `fiber_` has been added to the names of the existing, fiber-related objects.

6.2 Line of Count Support

The following features have been added to support line of count (LOC) for cables and cable segments:

- `mywcom_line_of_count`
- `mywcom_line_of_count_section`

To support the Line of Count functionality a new setting has been added. For details, see *Administrator GUI changes* on page 18

6.3 Explicit containment

New fields have been added to the technology-specific cable segment objects `mywcom_fiber_segment`, `mywcom_coax_segment` and `mywcom_copper_segment` to model explicit containment:

- **in_equipment**
- **out_equipment**

7: Upgrading

This section provides an overview of the upgrade process for this release.

7.1 Upgrading your database

To upgrade from the previous release of Network Manager Telecom:

1. Apply all IQGeo Platformupgrades:

   ```
   myw_db <db_name> upgrade core 70
   ```

2. Apply the Network Manager Telecom upgrade:

   ```
   myw_db <db_name> upgrade comms 7030
   ```

   The upgrade of the database has impact on these aspects:

   - It adds the Line of Count object and setting.
   - It adds the technology-specific system tables.
   - It adds the technology-specific layer groups.
   - It adds the technology designation prefix `fiber_` to existing features.

3. If any post-upgrade actions need to be performed, they are listed below.

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